LABORATORY ACTIVITY

Getting Familiar with Android Studio

I. Introduction

Before writing an Android application, a developer must first become familiar with the development environment. Android Studio contains several tools, panels, windows, and components that support the development, testing, debugging, and management of Android applications.

II. Learning Objectives

At the end of this activity, the student should be able to:

1. Launch and navigate the Android Studio development environment.
2. Identify the major components of the Android Studio user interface.
3. Recognize the basic tools and windows used during Android application development.
4. Create and open a basic Android project.
5. Identify the major folders and files in an Android project structure.
6. Explore the available tools for running, testing, and debugging an Android application.

III. Materials and Requirements

- Computer or laptop
- Installed Android Studio
- Android SDK
- Internet connection, if needed
- Android Emulator or connected Android device
- Activity sheet

ACTIVITY A: EXPLORE THE ANDROID STUDIO INTERFACE

Instructions

a. Open Android Studio.
b. Observe the main interface of the IDE.
c. Do not modify any project files yet.
d. Locate each component listed below.
e. Write the purpose or function of each component.

No.	Android Studio Component	Purpose / Function
1	Main Menu	
2	Toolbar	
3	Project Tool Window	
4	Editor Window	
5	Tool Windows	

6	Run Button	
7	Debug Button	
8	Device Selector	
9	Logcat	
10	Terminal	
11	Build Tool Window	
12	Problems Tool Window	

Guide Questions

a. Which area of Android Studio is used for writing and editing program files?
b. Which tool allows you to run an application?
c. Which tool is used to inspect application errors and debugging information?
d. Where can you view messages produced by the Android system and application?
e. Why is it important to become familiar with the IDE before writing code?

ACTIVITY B: CREATE AND EXPLORE A SAMPLE ANDROID PROJECT

Instructions

a. From Android Studio, select New Project.
b. Choose the appropriate project template as instructed by your instructor.
c. Configure the project using the following sample information:

Setting	Sample Information
Application Name	Hello Android
Language	Java
Minimum SDK	As instructed by the instructor

a. Finish creating the project.
b. Wait for Android Studio to complete the project setup.
c. Do not perform major code modifications yet.

Task 1: Identify the Project Structure

Explore the Project Tool Window and locate the following:

Component	Observed?	Purpose
Manifest Folder / File		
Java Source Files		
Resource Folder		

Layout Resources		
Drawable Resources		
Values Resources		
Gradle Scripts		

Task 2: Identify Important Project Files

Locate the following files and write their general purpose.

File / Component	General Purpose
AndroidManifest.xml	
Main Activity Java File	
Activity Layout XML File	
strings.xml	
colors.xml	
themes.xml	
Module-level Gradle File	

ACTIVITY C: EXPLORE THE DEVELOPMENT AND TESTING TOOLS

Without performing extensive coding, explore the following tools:

1. Device Manager

- What is its purpose?
- How can it be used during application testing?

2. Emulator

- What is an Android emulator?
- Why is it useful during application development?

3. Run Tool

- What happens when the Run button is pressed?

4. Debug Tool

- How is debugging different from simply running an application?

5. Logcat

- What type of information can be viewed in Logcat?

6. Build Tool

- What happens when an Android project is built?

ACTIVITY D: ANDROID STUDIO EXPLORATION CHALLENGE

Perform the following tasks.

[ ] Open Android Studio.
[ ] Create or open an Android project.
[ ] Locate the Project Tool Window.
[ ] Locate the main Java source file.
[ ] Locate the main XML layout file.
[ ] Locate the Android Manifest file.
[ ] Open Device Manager.
[ ] Check the available virtual devices.
[ ] Locate the Run button.
[ ] Locate the Debug button.
[ ] Open Logcat.
[ ] Open the Terminal window.
[ ] Open the Build tool window.
[ ] Identify the Problems tool window.
[ ] Locate the Gradle files.

ACTIVITY E: IDE FAMILIARIZATION MAP

Draw or paste a screenshot of the Android Studio interface and label at least eight (8) major components.

Your labels may include:

a. Main Menu
b. Toolbar
c. Project Tool Window
d. Editor Window
e. Device Selector
f. Run Button
g. Debug Button
h. Logcat
i. Terminal
j. Build Window

SPACE FOR ANDROID STUDIO SCREENSHOT OR INTERFACE MAP

ACTIVITY F: REFLECTION

Answer the following questions briefly.

a. Which Android Studio component did you find most useful? Why?
b. Which component was initially difficult for you to understand?
c. Why should a developer understand the project structure?
d. How can familiarization with the IDE help make coding easier?
e. What do you expect to learn when you begin actual Android application coding?

IV. Expected Output

At the end of the activity, students should submit:

a. A completed Android Studio interface identification table.
b. A completed Android project structure table.
c. Answers to the guide questions.
d. A labeled Android Studio interface map or screenshot.
e. A brief reflection.

V. Assessment Rubric

Criteria	Excellent	Satisfactory	Needs Improvement
Interface Identification	Correctly identifies all or nearly all components	Identifies most components	Identifies only a few components
Project Structure Exploration	Demonstrates clear understanding of project folders and files	Demonstrates partial understanding	Has difficulty identifying project components
Completion of Tasks	Completes all required tasks	Completes most tasks	Completes only a few tasks
Reflection	Shows clear understanding and meaningful insights	Shows adequate understanding	Shows limited understanding
Overall Output	Complete, organized, and accurate	Mostly complete and organized	Incomplete or poorly organized